

	- the question requires the application of Faraday's law from the moment the battery is connected.	
7(a)(i)	13.6 eV	[1] for correct answer
7(a)(ii)	$\text{KE of electrons} = \frac{2.00 \times 10^{-18}}{1.60 \times 10^{-19}}$ $= 12.5 \text{ eV}$ Hence the highest state that the atom can be excited to is n=3. Only the following transitions are allowed From n =3 to n=2 From n =2 to n=1 From n =3 to n=1 Hence number of different wavelengths of radiation that can be emitted = 3	[1] for correct highest state [1] for correct answer
7(a)(iii)	Zero. The energy of the photon does not match exactly any of the energies required for the atom to get excited from the ground state to higher excited state hence the photons incident will not be absorbed by the atom.	[1] [1]

7(b)(i)	From the graph, minimum wavelength $\lambda_{\min}$ is the same for both spectra.	[1]
7(b)(ii)	KE lost by electron = energy of emitted X-ray photon $eV = \frac{hc}{\lambda_{\min}}$ $\lambda_{\min} = 16 \times 10^{-12} \text{ m}$ $V = \frac{hc}{e\lambda_{\min}} = \frac{6.63 \times 10^{-34} \times 3.00 \times 10^8}{1.60 \times 10^{-19} \times 16 \times 10^{-12}}$ $= 7.77 \times 10^4 \text{ V}$	[1] for correct substitution [1] for correct answer
8(a)	The velocity changes as the direction of motion changes with time	[1]